1.04 Graph Sketching: Linear Graphs

Chapter 5 · Original slide 4 · Let's start simple…What is the equation of this line?And more importantly, why is it that?



Let's start simple...

What is the equation of this line?

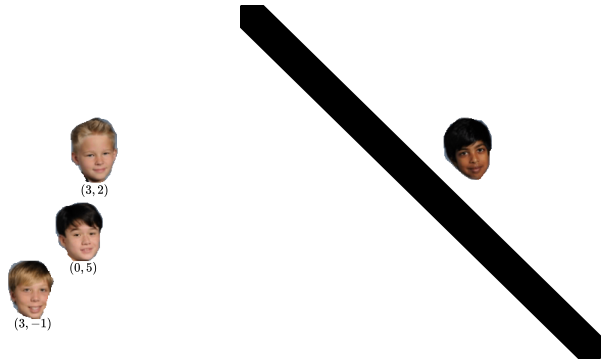
And more importantly, why is it that?

4
3
2
1
-1
-2
-3
-4

x -5 -4 -3 -2 -1 0 1 2 3 4 5 6

For any point we pick on the line, the x value is always 2.

Chapter 5 · Original slide 5 · Lines and Equations of Lines



! A line consists of all points which satisfy some equation in terms of x and/or y.

Anirudh's Line Club

Membership Rules:

$2x + y = 5$

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Chapter 5 · Original slide 6 · Examples

This means we can substitute the values of a coordinate into our equation whenever we know the point lies on the line.

Find the coordinate of the point where the line $2x + y = 5$ cuts the x-axis.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of a.

Chapter 5 · Original slide 7 · Test Your Understanding

Determine where the line $x + 2y = 3$ crosses the:

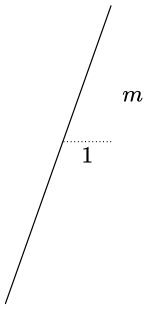
a) y-axis:

b) x-axis:

What mistakes do you think it's easy to make?

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Chapter 5 · Original slide 8 · Recap of gradient



The steepness of a line is known as the gradient.

It tells us what y changes by as x increases by 1.

So if the y value increased by 6 as the x value increased by 2, what is y increasing by for each unit increase of x?

How would that give us a suitable formula for the gradient m?

$m = \frac{\Delta y}{\Delta x}$

Δ is the (capital) Greek letter "delta" and means "change in".

Textbook Note:

The textbook uses $m = \frac{y_2 - y_1}{x_2 - x_1}$ for two points (x_1, y_1) and (x_2, y_2).

Reasons I don't use it for non-algebraic coordinates:

• Students often get the y_1 and y_2 the wrong way round (or with the x's)

• Students often make sign errors when dealing with negatives, e.g. $(-3) - (-4)$

• It can't be done as easily mentally,

• Students see it as "yet another formula to learn" when really all you need is to appreciate is what gradient is, i.e. "y change per x change".

Chapter 5 · Original slide 9 · Examples

Find the gradient of the line that goes through the points:

1

$(1, 4)(3, 10)$

2

$(5, 7)(8, 1)$

3

$(2, 2)(-1, 10)$

4

Show that the points $\mathrm{A}(3, 4), B(5, 5), C(11, 8)$ all lie on a straight line.

If points are 'collinear' they lie on the same line.

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Chapter 5 · Original slide 10 · Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1.
Work out the value of a.

Chapter 5 · Original slide 11 · y=mx+c

One form we can put a straight line equation in is:

$y = mx + c$

Gradient

y-intercept

Why does it work?

- The $\mathbf{y}$-intercept by definition is the $\mathbf{y}$ value when $\mathbf{x=0}$. Substituting:
$\mathbf{y=(m\times 0)+c=c}$ as expected.
- By the definition of gradient, if we increase $\mathbf{x}$ by 1, $\mathbf{y}$ should increase by $\mathbf{m}$:
$\mathbf{y=m\,(x+1)+c=mx+m+c}$
which indeed has increased by $\mathbf{m}$.

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Chapter 5 · Original slide 12 · y=mx+c

Determine the gradient and y-intercept of the line with equation $4x - 3y + 5 = 0$

Make y the subject so we have the form

$y = mx + c$

Put y on the side it's positive.

Divide each term by 3; don't write $y = \frac{4x+5}{3}$ otherwise it's not in the form $y = mx + c$

This is algebra, so use improper fractions, and not mixed numbers or recurring decimals.

Chapter 5 · Original slide 13 · ax+by+c=0

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the 'slope-intercept form'.

But another common form is $\mathbf{ax+by+c=0}$, where a, b, c are integers. This is known as the 'standard' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where a, b, c are integers.

We don't want fractions, so multiply by an appropriate number.

Put everything on either side of equation. $-x + 3y + 2 = 0$ would also be OK.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$...

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Chapter 5 · Original slide 14 · Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$2x + y = 4$



$y = mx + c$

“Slope-Intercept Form”

$ax + by + c = 0$

“Standard Form”

Coverage

$y = mx + c$ doesn’t allow you to represent vertical lines. Standard form allows us to do this by just making b zero.

Usefulness

This more ‘elegant’ form also means it ties in with vectors and matrices. In FM, you will learn about the ‘dot product’ of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

thus since $ax + by + c = 0$, we can represent a straight line using:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 (1)$$

Symmetry

In general, the ‘linear combination’ of two variables x and y is $ax + by$, i.e. “some amount of x and some amount of y”. There is a greater elegance and symmetry to this form over $y = mx + c$ because x and y appear similarly within the expression.

We can extend to 3D points to get the equation of a plane:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 (2)$$

Conveniently, in equation (1), the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is perpendicular to the line. And in equation (2), the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$x + 4 = 0$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$$

$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Chapter 5 · Original slide 15 · Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where a, b, c are integers.

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Chapter 5 · Original slide 16 · Exercise 5A – Pure Y1 Page 91

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. (3 marks)

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

Chapter 5 · Original slide 17 · Exercise 5B – Pure Y1 Page 92-93

8 A line l cuts the x-axis at $(5, 0)$ and the y-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

10 The line l with gradient 3 and y-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

Find the values of a and c. (2 marks)

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the y-axis at $-\frac{c}{b}$.

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Chapter 5 · Original slide 18 · Exercise 5B – Pure Y1 Page 92-93

11 The straight line l passes through $(0, 6)$ and has gradient -2. It intersects the line with equation $5x - 8y - 15 = 0$ at point P. Find the coordinates of P.

12 The straight line l_1 with equation $y = 3x - 7$ intersects the straight line l_2 with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of b.

b Find the value of a.